

Dr. Sagar Hosangadi Prutvi



Experience

Postdoc Researcher

Albert-Ludwigs-Universität Freiburg

May, 2024 - Present

- Developed an ML-based system that uses force sensor data to implement surface parallelization in tribo devices.
- Built Python-based automation tools for multi-device lab instrumentation and data acquisition.
- Designed a frequency tunable triboelectric energy harvester.

Data & Analytics specialist

SwissRe

Dec, 2023 - Apr, 2024

- Developed a full-stack client-ranking ML pipeline, reducing audit prep time by 90% across global teams, implemented using PySpark, SQL, PowerBI and Palantir Foundry.

Data Scientist

Halliburton

Jun, 2021 - Aug, 2023

- Proposed and submitted two patent ideas involving novel sensor configurations for harsh industrial environments.
- Led an AI/ML initiatives & built a digital solution to predict the application runtime with >95% accuracy (E2E delivery).

Mechanical (CAD) Engineer

HCL Technologies

Sep, 2011 - Aug, 2013

- Providing CAD Support for product development activities of Ion Implanters (client: Varian Semiconductor Equipments).

Education

Doctor of Philosophy (Ph. D.) | Sensors and μ -energy harvesters

Indian Institute of Technology Bombay.

2016 – 2022

- Thesis: Triboelectric effect driven self-powered vibration sensors and wind energy harvesting device for enabling Industry 4.0.

Master of Technology (M. Tech) | Nanotechnology

National Institute of Technology Karnataka Surathkal

2013 – 2015

- Thesis: Design and analysis of 1D silicon photonic crystal-based strain and mass sensor.

Bachelor of Engineering (B.E.) | Mechanical Engineering

Visvesvaraya Technological University.

2007 – 2011

Summary

Experienced Data Scientist and Engineering Researcher with 7+ years in industry, applying machine learning to sensor-based systems, diagnostics, and process optimization. Skilled in building end-to-end ML pipelines, cloud-based data workflows (Azure, AWS), and Power BI dashboards. Strong foundation in physics and materials science with hands-on exposure to semiconductor equipment, silicon wafer processing, and cleanroom practices. Adept at bridging experimental systems with data modeling to drive digital transformation in high-tech environments.

Contact

Phone:

(+49) 15560 218813

E-Mail:

sagarhp3589@gmail.com

Address:

Sundgaualle 58,
79110 Freiburg, Germany

Linkedin:

linkedin.com/in/hpsagar

E-portfolio:

hprutvisagar.github.io

Github:

github.com/hprutvisagar

Skills

- **Machine Learning & Data Tools:**
Python | SQL | Scikit-learn | TensorFlow | Keras | Mlflow | PySpark | Power BI | MATLAB
- **Cloud Platforms:**
Azure (incl. ML & Data Factory) | AWS (SageMaker, Canvas) | Palantir foundry (ontology)
- **Data Engineering & Modeling:**
End-to-end ML Pipelines | Anomaly Detection | Feature Engineering | Data Cleaning, wrangling & Governance
- **Domain & Simulation Tools:**
COMSOL | Ansys | Solidworks | AutoCAD | Silicon Wafer Processing | Electro-mechanics | Cleanroom Protocols

Certifications

- Generative AI with Large Language Models (Coursera, 2025)
- Azure Machine Learning & MLOps: Beginner to Advance (Udemy, Ongoing)
- Natural Language Processing with Python (Udemy, 2022)
- Python for Time Series Data Analysis (Udemy, 2022)
- Python for Data Science and Machine Learning bootcamp (Udemy, 2021)

Selected publications

- **Self-powering vibration sensor based on a cantilever system with a single electrode mode triboelectric nanogenerator.**
Sagar Hosangadi Prutvi, Mallikarjuna Korrapati, et al.
Measurement Science and Technology, 33 (7), 075115 (2022)
- **Triboelectric effect based self-powered compact vibration sensor for predictive maintenance of industrial machineries.**
Sagar Hosangadi Prutvi, Sunil Meti, et al.
Measurement Science and Technology, 32 (9), 095119 (2021)
- **Transient dynamic distributed strain sensing using photonic crystal waveguides.**
Sagar Hosangadi Prutvi, Vignesh Mahalingam, et al.
Applied Optics, 56 (28), 7877-7885 (2017)
- **Graphene Integrated Waveguide for Molecular Sensing.**
Sagar Hosangadi Prutvi, and MR Rahman.
International Engineering Symposium (IES), Kumamoto University, Japan (2015)

Languages

- **English** : Professional
- **German** : A2
(Actively upskilling)
- **Kannada** : Native speaker

Interests

- Sensor Systems & Condition Monitoring
- Predictive Modeling & Process Optimization
- AI for Manufacturing & Diagnostics
- Semiconductor Technology & Materials Engineering
- Systems Thinking & Interdisciplinary Innovation
- Hiking, Table Tennis, and Outdoor Activities